

will provide significantly brighter viewfinder images while also giving you faster and more accurate auto-focusing in such scenes. Minimum aperture lenses find very limited need because of the photo blurring from lens diffraction. Mostly in cases when extreme depth of field is desired, smaller apertures and larger f-numbers find use. Also some zoom lenses on dSLRs and even compact digital cameras give you a list of maximum apertures depending on how far you wish to zoom in or out, thus it's important to note that aperture ranges mentioned here refer only to the range of maximum aperture rather than the overall range.

For the sake of completeness, we must mention that an aperture opening (iris) is rarely a perfect circle, due to the presence of 5-8 blade-like lens diaphragms.

Shutter speed

This indicates the time that the shutter (internal cover of lens) opens up to let the light reach the sensor. Higher the shutter speed, smaller is the duration and lesser the light that reaches the sensor (relate to the time of contact of sketch pen and paper). It simply indicates how long the incident light will be permitted to enter the camera and hit the sensor. Note that shutter speed and exposure time are conceptually the same, with longer exposure time implying slower shutter speed. Not only does it affect exposure but variations in shutter speed can change the way movement appears in the shot. For instance, at sporting events to freeze fast-moving subjects, very short shutter speeds are beneficial. On the other hand, if you want a more artistic effect by introducing additional motion blur in moving subjects like in a waterfall, longer shutter speeds give you beautiful shots.



Extremely high shutter speed to capture half burned head of a matchstick

Courtesy: Subiet@Flickr